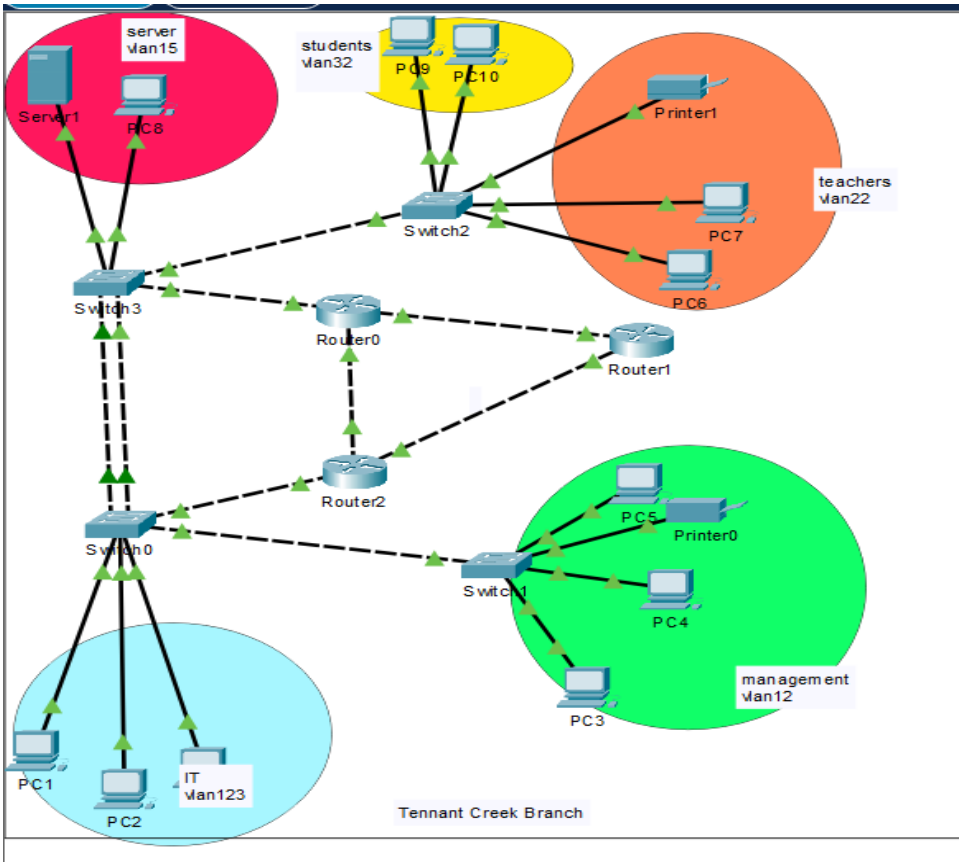
Tennent Creek Branch:

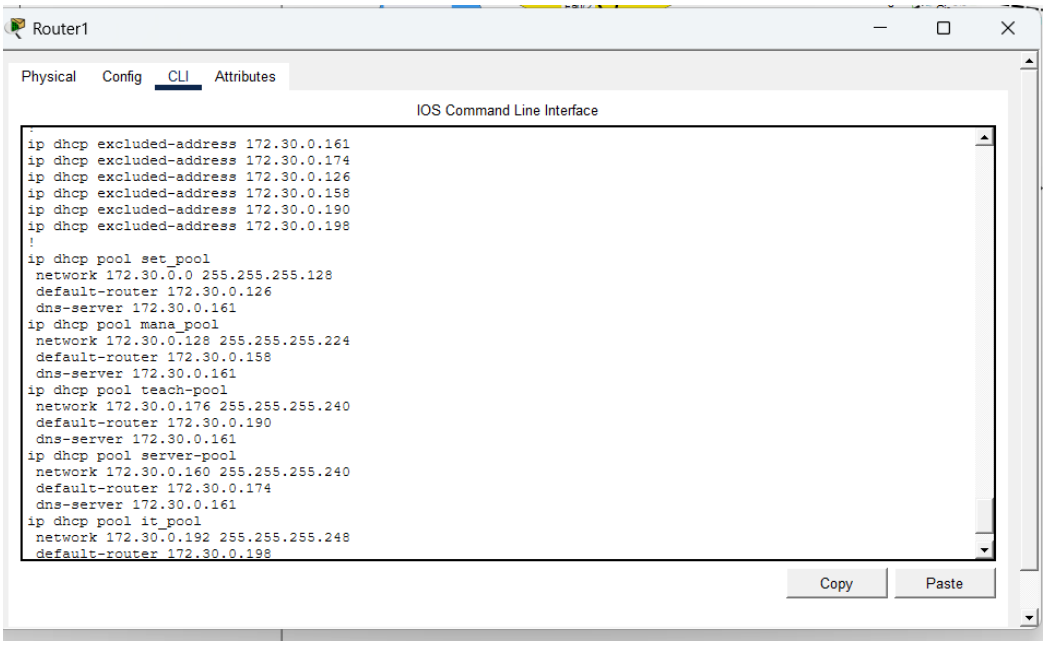


The Tennant Creek branch network was designed as part of a multi-site school infrastructure based in Darwin, supporting centralized services and scalable connectivity. The branch utilizes the `172.30.0.0/22` address space, with IP allocation performed using VLSM based on departmental requirements.

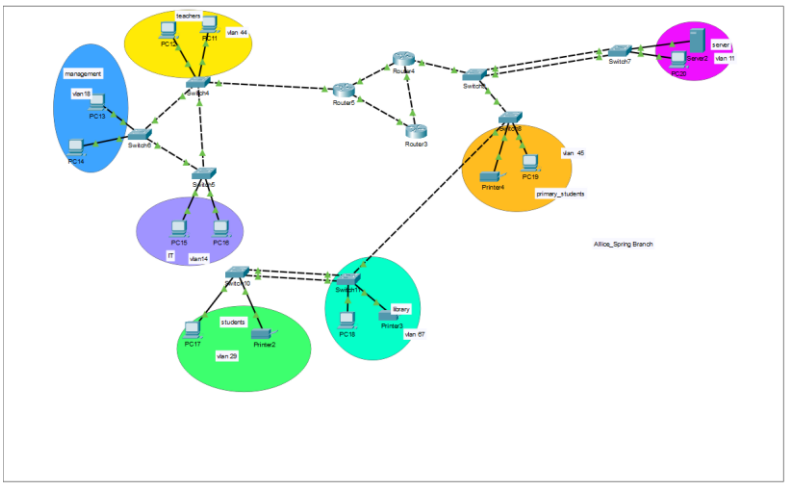
VLAN segmentation was implemented to separate network traffic across different user groups, including Students (80 users), Management (15 users), Teachers (9 users), IT (3 users), and Servers (10 devices). Each VLAN was assigned an appropriate subnet, such as `172.30.0.0/25` for students and `172.30.0.128/27` for management, ensuring efficient IP utilization and future scalability.

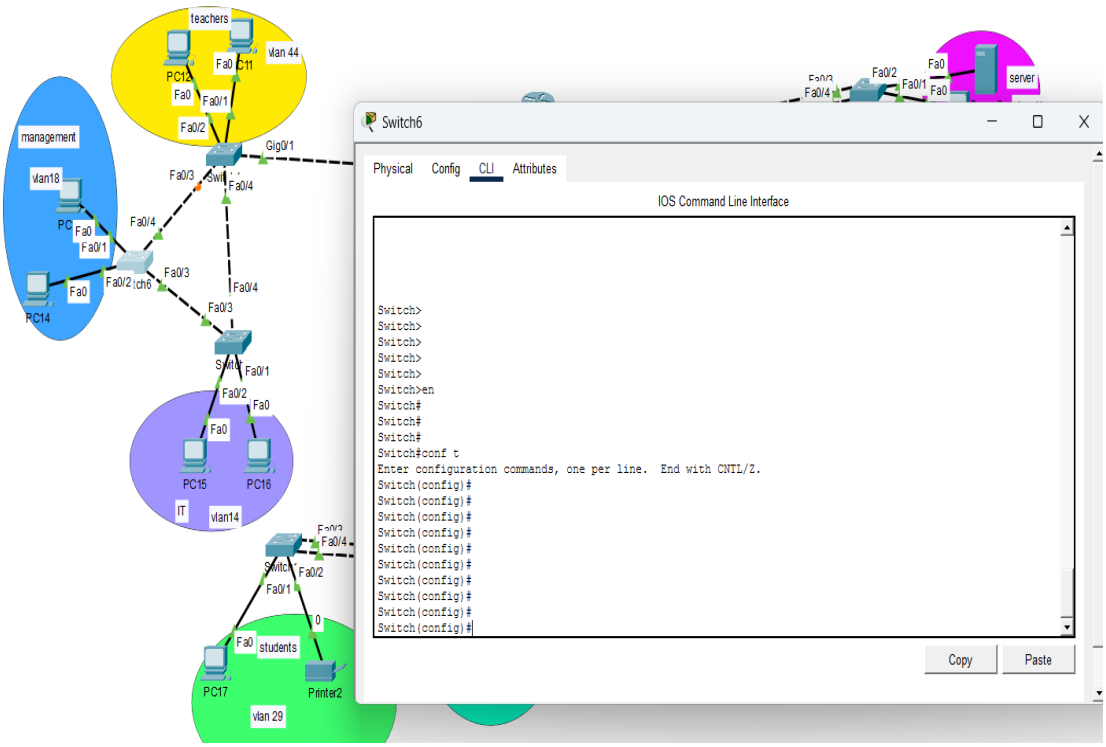
Inter-VLAN routing was achieved using a router-on-a-stick configuration, with default gateways configured on router subinterfaces. Centralized DHCP services were provided by a remote router, while the local router at the branch acted as a DHCP relay agent using the `ip helper-address` feature.

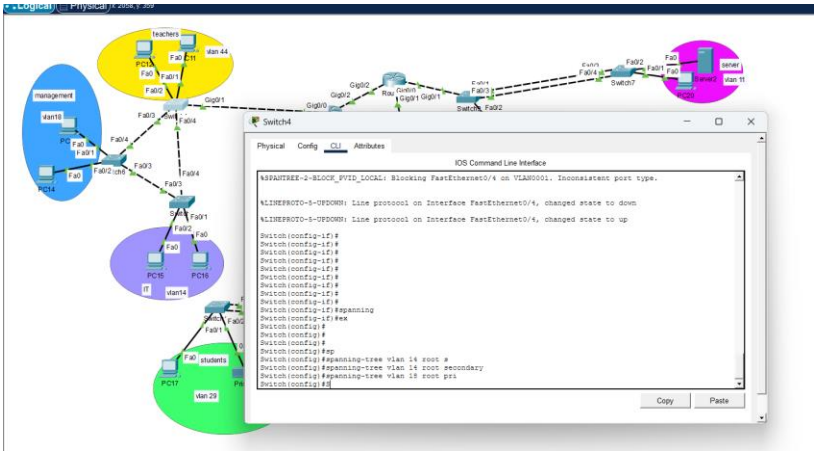
During testing, an issue was encountered where DHCP requests were not being fulfilled despite correct VLAN configuration, trunking, and routing. The problem was identified as incorrect placement of the DHCP relay configuration on the physical interface instead of the VLAN subinterfaces. After correcting this, DHCP communication was successfully restored across all VLANs.



Allice Spring Branch:




teachers
vlan 44
PC12
PC11
Fa0
Fa0/1
Fa0/2
management
vlan18
PC13
PC14
Gig0/1
Fa0/3
Fa0/4
Fa0/2
Fa0/3
Fa0/4
Fa0/3
Fa0/1
Fa0/2
Fa0
PC15
PC16
IT
vlan14
Fa0/4
Fa0/2
Fa0/1
Fa0
students
PC17
Printer2
vlan 29
server
Switch6
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>en
Switch#
Switch#
Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Copy
Paste



The switching infrastructure for the Alice Springs branch was designed to provide redundancy and efficient traffic distribution. Multiple switches were interconnected using EtherChannel links to ensure high availability and increased bandwidth between network segments.

Spanning Tree Protocol (STP) was implemented to prevent Layer 2 loops within the network. To optimize traffic flow, specific switches were configured as the root bridge for selected VLANs using the `spanning-tree vlan <id> root primary` command. This allowed different VLANs to utilize different forwarding paths, effectively achieving load balancing across redundant links while maintaining network stability.

The branch network was assigned the `172.20.0.0/22` address space. Variable Length Subnet Masking (VLSM) was applied to allocate IP addresses efficiently based on the number of devices in each department.

The subnet allocation is as follows:

- Students (130 users): `172.20.0.0/24`
- Primary Students (35 users): `172.20.2.0/26`
- Teachers (25 users): `172.20.2.64/27`
- Management (19 users): `172.20.2.96/27`
- Servers (8 devices): `172.20.2.128/28`
- Library (5 devices): `172.20.2.144/29`
- IT (4 devices): `172.20.2.152/29`

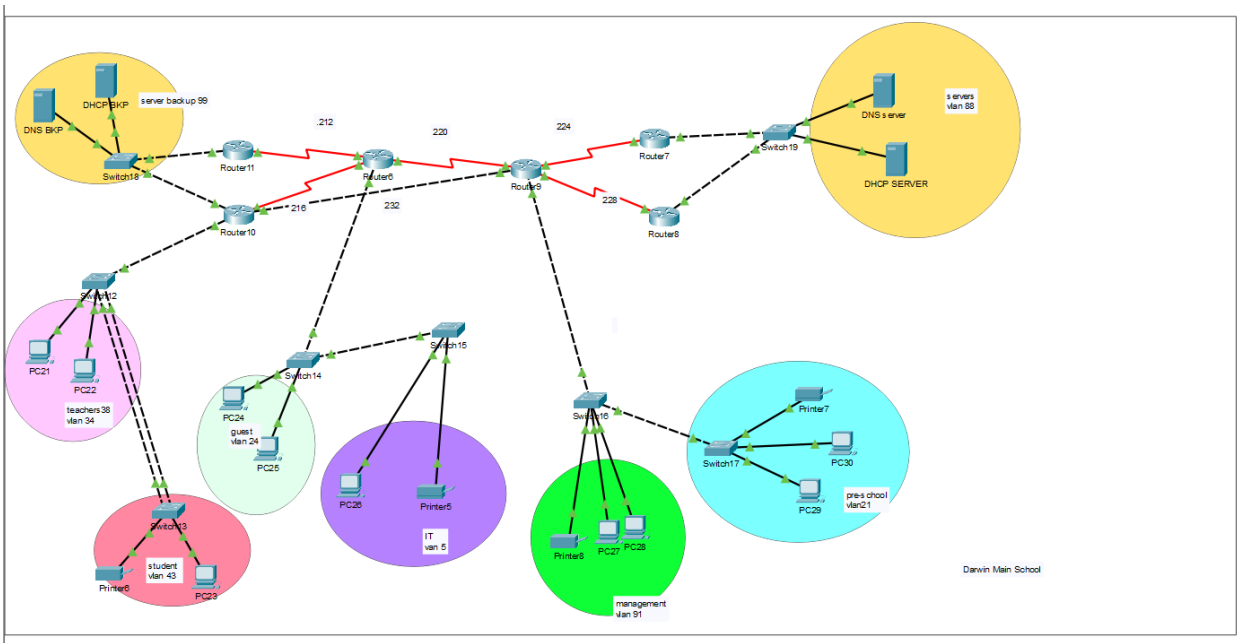
Each subnet was assigned as a default gateway configured on router subinterfaces, enabling inter-VLAN communication through a router-on-a-stick architecture. During testing, devices in the student VLAN experienced connectivity issues, including failure

to obtain IP addresses via DHCP and inability to reach the default gateway. This occurred despite correct VLAN configuration, trunking, and EtherChannel setup.

Upon further investigation, the issue was identified as an IP addressing mismatch. The default gateway for VLAN 29 was configured in a different subnet (`172.20.1.254/24`) than the client devices (`172.20.0.0/24`). This prevented proper communication, as hosts were unable to resolve the gateway address.

The issue was resolved by correcting the gateway configuration to match the appropriate subnet (`172.20.0.254/24`). After this change, normal network functionality was restored, and DHCP services operated as expected.

Darwin Main School:

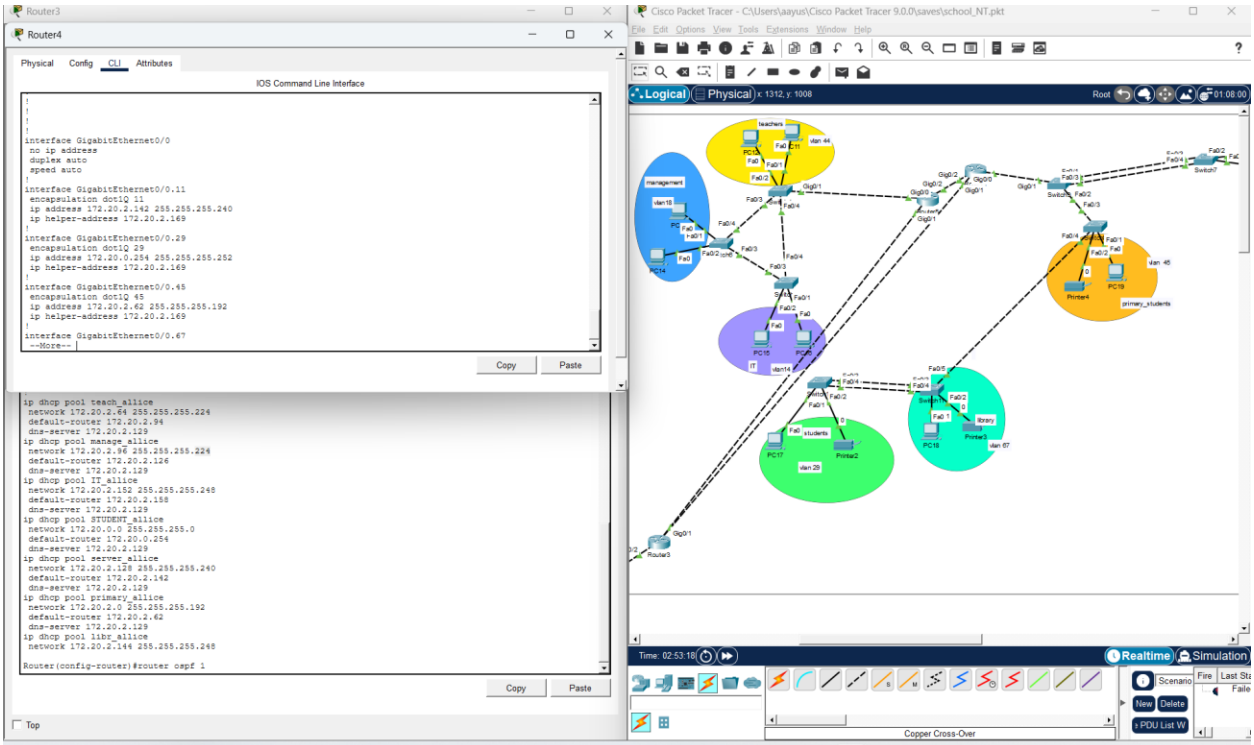


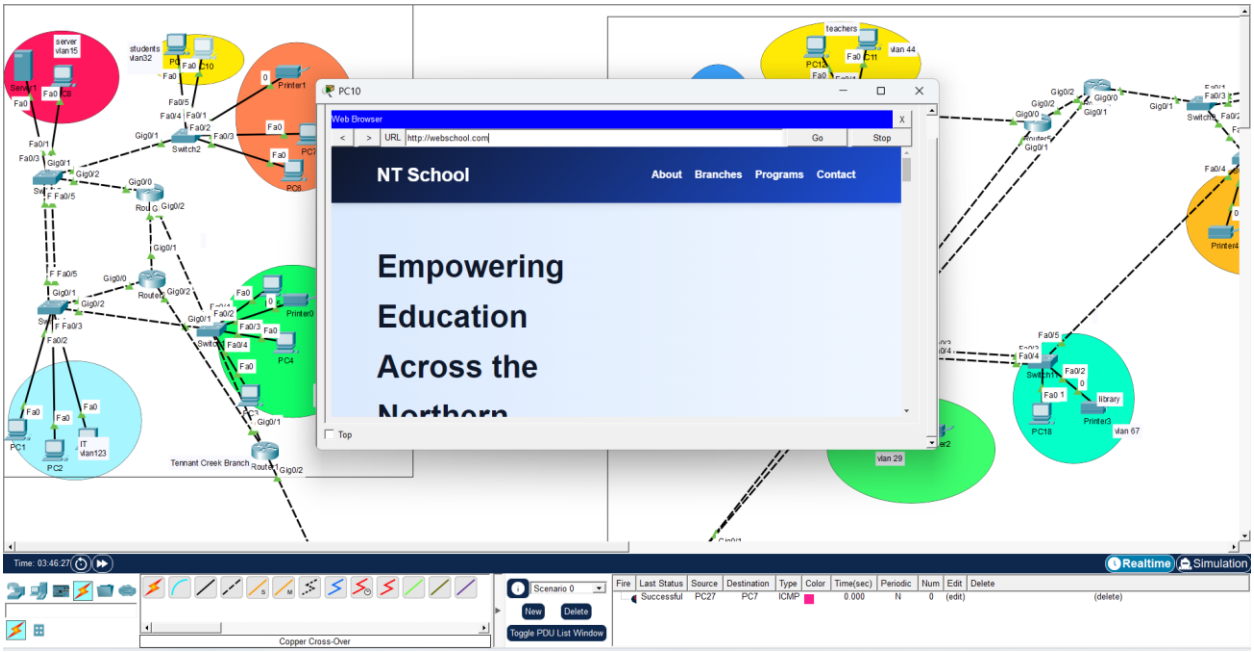
Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#ip address 203.10.10.5 255.255.255.252
Router(config-if)#int g0/2
Router(config-if)#ip address 203.10.10.9 255.255.255.252
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#
Router#
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 203.10.10.0 0.0.0.3 area 0
Router(config-router)#network 203.10.10.4 0.0.0.3 area 1
Router(config-router)#
```

Copy Paste


Router3
Router4
Physical
Config
CLI
Attributes
IOS Command Line Interface
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
interface GigabitEthernet0/0.11
encapsulation dot1Q 11
ip address 172.20.2.142 255.255.255.240
ip helper-address 172.20.2.169
interface GigabitEthernet0/0.29
encapsulation dot1Q 29
ip address 172.20.0.254 255.255.255.252
ip helper-address 172.20.2.169
interface GigabitEthernet0/0.45
encapsulation dot1Q 45
ip address 172.20.2.62 255.255.255.192
ip helper-address 172.20.2.169
interface GigabitEthernet0/0.67
--More--
Copy
Paste
ip dhcp pool teach_allice
network 172.20.2.64 255.255.255.224
default-router 172.20.2.94
dns-server 172.20.2.129
ip dhcp pool manage_allice
network 172.20.2.96 255.255.255.224
default-router 172.20.2.126
dns-server 172.20.2.129
ip dhcp pool IT_allice
network 172.20.2.152 255.255.255.248
default-router 172.20.2.158
dns-server 172.20.2.129
ip dhcp pool STUDENT_allice
network 172.20.0.0 255.255.255.0
default-router 172.20.0.254
dns-server 172.20.2.129
ip dhcp pool server_allice
network 172.20.2.128 255.255.255.240
default-router 172.20.2.142
dns-server 172.20.2.129
ip dhcp pool primary_allice
network 172.20.2.0 255.255.255.192
default-router 172.20.2.62
dns-server 172.20.2.129
ip dhcp pool libr_allice
network 172.20.2.144 255.255.255.248
Router(config-router)#router ospf 1
Copy
Paste
Top
Cisco Packet Tracer - C:\Users\aayus\Cisco Packet Tracer 9.0.0\saves\school_NT.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
Time: 02:53:18
Realtime
Simulation
Copper Cross-Over

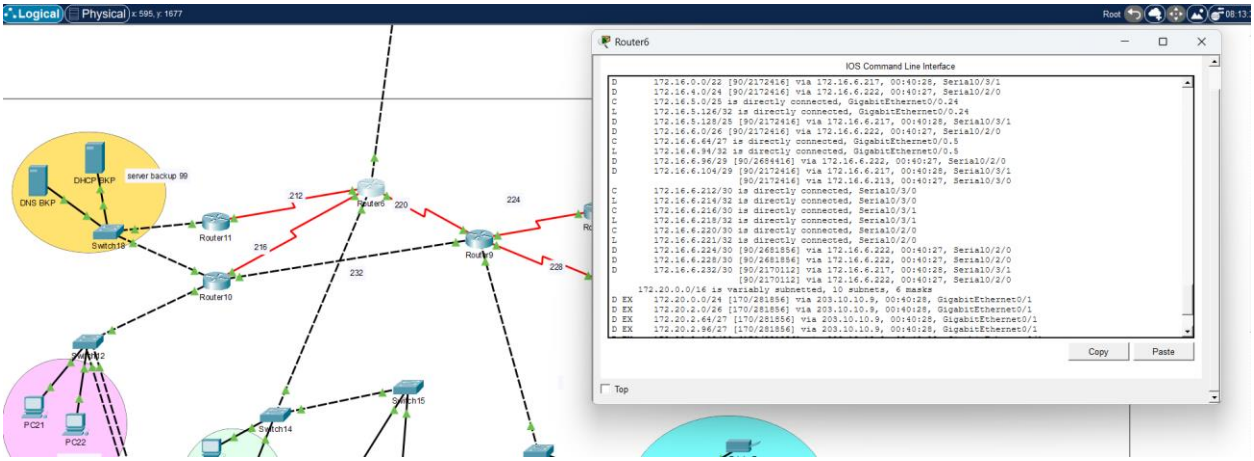

PC10
Web Browser
URL http://webschool.com
Go
Stop
NT School
About
Branches
Programs
Contact
Empowering
Education
Across the
Northern
Top
Time: 03:46:27
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Successful
PC27
PC7
ICMP
0.000
N
0
(edit)
(delete)
Copper Cross-Over

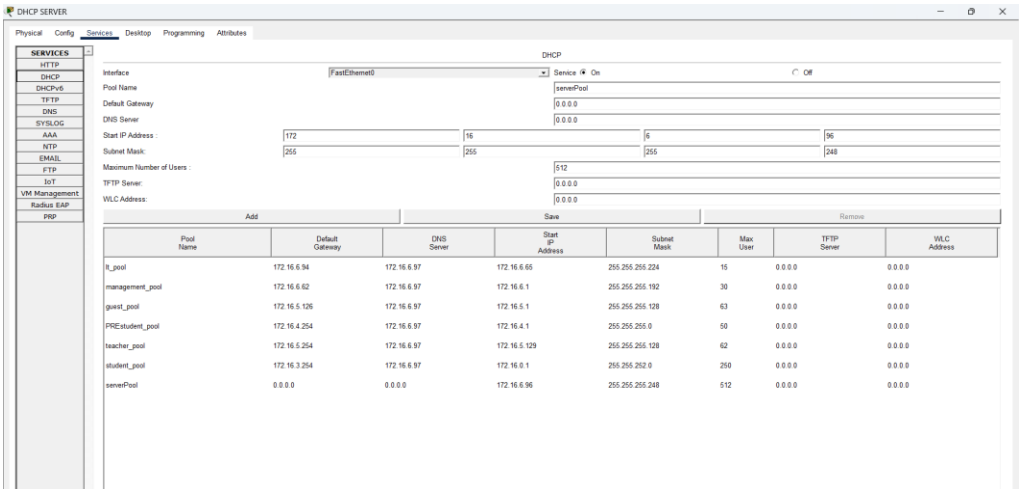
PC10

Command Prompt

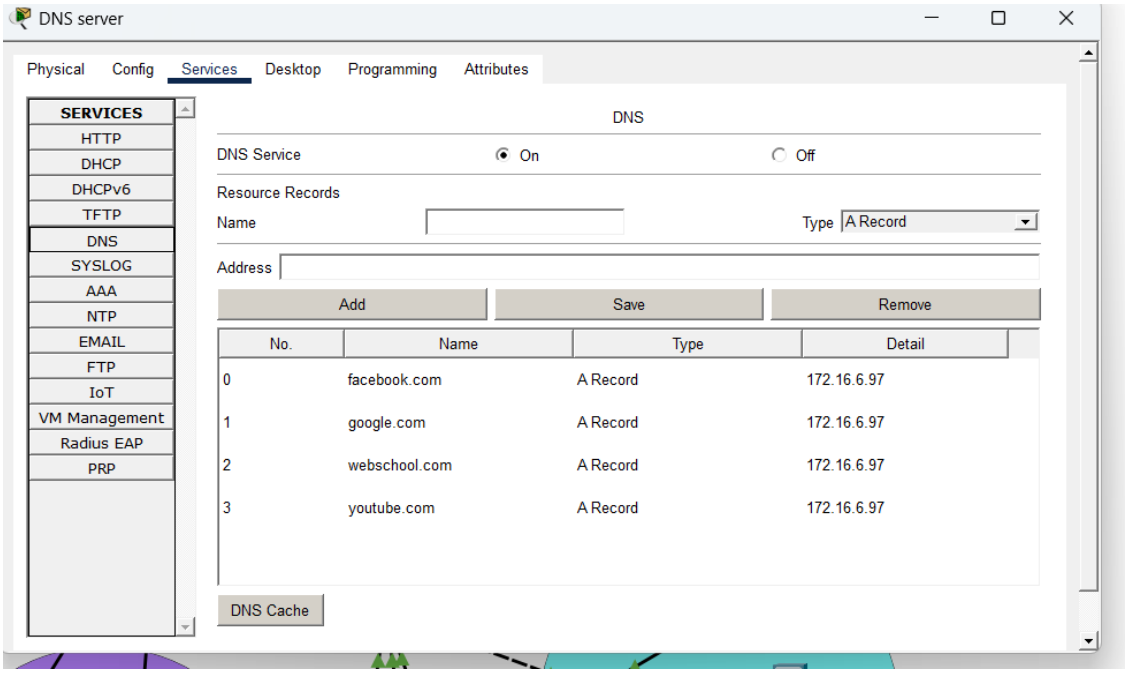
```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.20.2.68

Pinging 172.20.2.68 with 32 bytes of data:

Reply from 172.20.2.68: bytes=32 time<1ms TTL=123
Reply from 172.20.2.68: bytes=32 time<1ms TTL=123
Reply from 172.20.2.68: bytes=32 time<1ms TTL=123
Reply from 172.20.2.68: bytes=32 time<1ms TTL=123

Ping statistics for 172.20.2.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.20.2.68

Pinging 172.20.2.68 with 32 bytes of data:

Reply from 172.30.0.206: Destination host unreachable.
Reply from 172.30.0.206: Destination host unreachable.
Reply from 172.30.0.206: Destination host unreachable.
Reply from 172.30.0.206: Destination host unreachable.

Ping statistics for 172.20.2.68:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

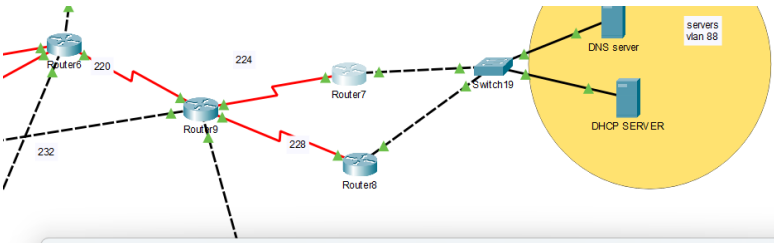
Top





Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
lt_pool	172.16.6.94	172.16.6.97	172.16.6.65	255.255.255.224	15	0.0.0.0	0.0.0.0
management_pool	172.16.6.62	172.16.6.97	172.16.6.1	255.255.255.192	30	0.0.0.0	0.0.0.0
guest_pool	172.16.5.126	172.16.6.97	172.16.5.1	255.255.255.128	63	0.0.0.0	0.0.0.0
PREstudent_pool	172.16.4.254	172.16.6.97	172.16.4.1	255.255.255.0	50	0.0.0.0	0.0.0.0
teacher_pool	172.16.5.254	172.16.6.97	172.16.5.129	255.255.255.128	62	0.0.0.0	0.0.0.0
student_pool	172.16.3.254	172.16.6.97	172.16.0.1	255.255.252.0	250	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.6.96	255.255.255.248	512	0.0.0.0	0.0.0.0


DNS server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
facebook.com
A Record
172.16.6.97
1
google.com
A Record
172.16.6.97
2
webschool.com
A Record
172.16.6.97
3
youtube.com
A Record
172.16.6.97
DNS Cache

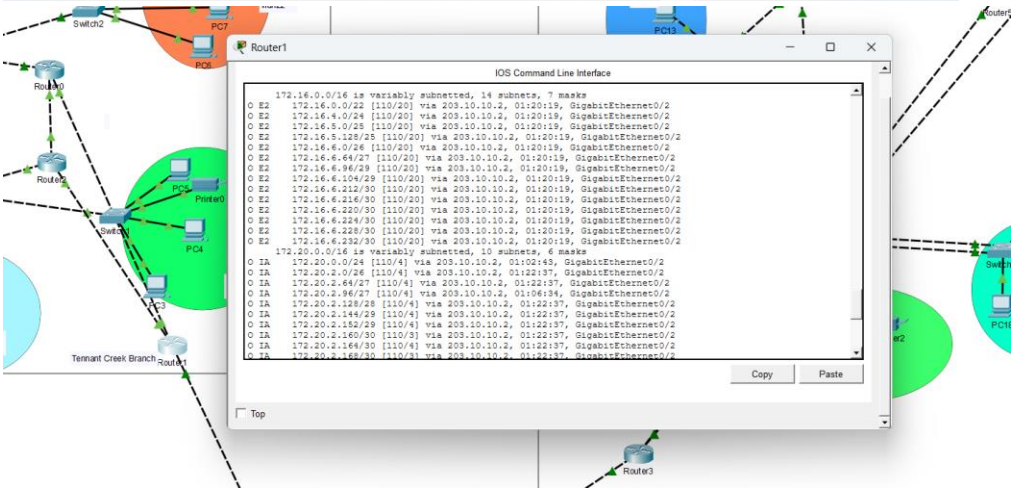


Router7

IOS Command Line Interface

```
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#do sh standby
GigabitEthernet0/0.88 - Group 1 (version 2)
  State is Active
    7 state changes, last state change 00:18:40
  Virtual IP address is 172.16.6.99
  Active virtual MAC address is 0000.0C9F.F001
    Local virtual MAC address is 0000.0C9F.F001 (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.696 secs
  Preemption disabled
  Active router is local
  Standby router is 172.16.6.101
  Priority 100 (default 100)
  Group name is hsrp-Gig-1 (default)
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#
```

Copy Paste



Router1

IOS Command Line Interface

```
     172.16.0.0/16 is variably subnetted, 14 subnets, 7 masks
O E2    172.16.0.0/22 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.4.0/24 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.5.0/25 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.5.128/25 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.0/26 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.64/27 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.96/29 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.104/29 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.212/30 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.216/30 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.220/30 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.224/30 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.228/30 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
O E2    172.16.6.232/30 [110/20] via 203.10.10.2, 01:20:19, GigabitEthernet0/2
     172.20.0.0/16 is variably subnetted, 10 subnets, 6 masks
O IA    172.20.0.0/24 [110/4] via 203.10.10.2, 01:02:43, GigabitEthernet0/2
O IA    172.20.2.0/26 [110/4] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
O IA    172.20.2.64/27 [110/4] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
O IA    172.20.2.96/27 [110/4] via 203.10.10.2, 01:06:34, GigabitEthernet0/2
O IA    172.20.2.128/28 [110/4] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
O IA    172.20.2.144/29 [110/4] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
O IA    172.20.2.152/29 [110/4] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
O IA    172.20.2.160/30 [110/3] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
O IA    172.20.2.164/30 [110/4] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
O IA    172.20.2.168/30 [110/3] via 203.10.10.2, 01:22:37, GigabitEthernet0/2
```

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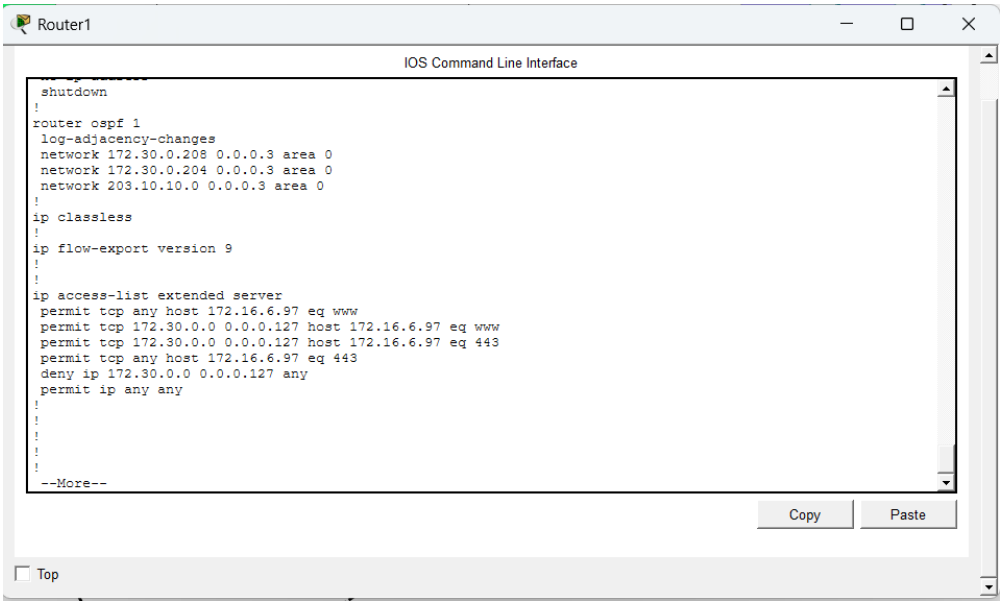
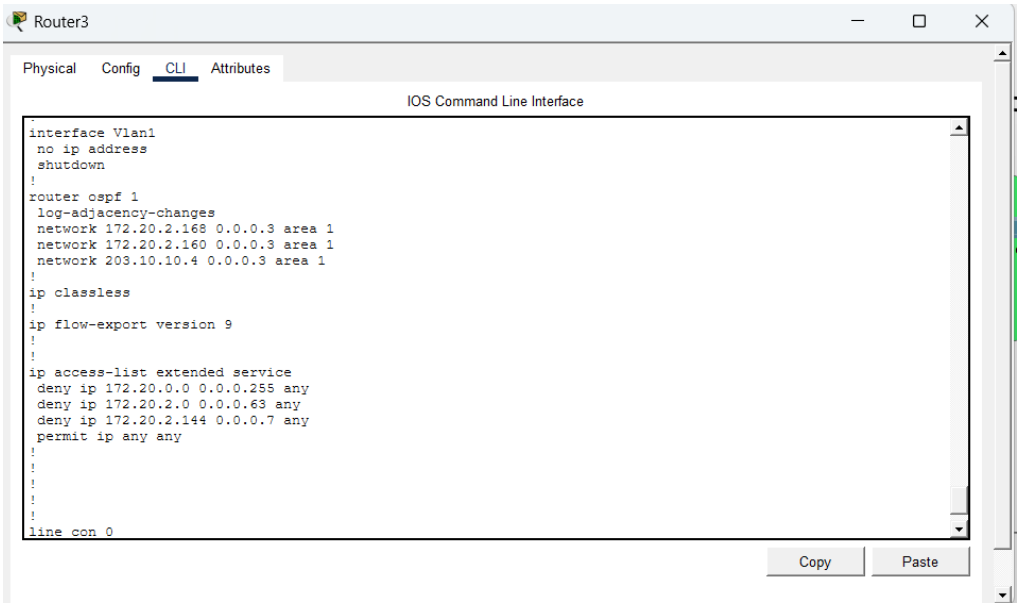
Top





The Darwin site functions as the central hub of the network, hosting critical services and acting as the backbone for inter-branch communication. It is responsible for providing centralized resources such as DNS, web services, and DHCP support to all connected branches, including Tennant Creek and Alice Springs.

The network at Darwin is configured using the EIGRP routing protocol, which ensures efficient and scalable routing within the core domain. To enable communication with remote branches running OSPF (Area 0 and Area 1), route redistribution is implemented on the boundary router, allowing seamless integration between different routing protocols.

Darwin also hosts the primary DNS and HTTP server, which provides domain name resolution and web services (e.g., *school.com*) accessible from all branches. This centralized approach ensures consistency in service delivery and simplifies management.

High availability is achieved through the implementation of HSRP (Hot Standby Router Protocol), which provides gateway redundancy and ensures uninterrupted network access in case of device failure. Additionally, DHCP services are configured to dynamically assign IP addresses, default gateways, and DNS settings to client devices across multiple VLANs.

To enhance network security, Access Control Lists (ACLs) are deployed to regulate traffic flow. These policies ensure that users, such as students, can access required services like the web server while restricting unauthorized or unnecessary communication.

Overall, the Darwin Main School network serves as a robust and scalable core infrastructure, integrating routing, redundancy, centralized services, and security mechanisms to support reliable communication across all branches.

Problem Encountered:

During the implementation of the network, several issues were encountered, particularly while configuring routing, addressing, and services across multiple branches. One of the initial problems involved incorrect subnet masking within VLAN configurations, which prevented devices from communicating with their default gateway. This was resolved by verifying and correcting the subnet mask to match the designed IP addressing scheme.

Another issue arose during the configuration of HSRP, where duplicate IP address errors occurred due to misconfiguration of the virtual IP and interface IPs. This was fixed by ensuring that the HSRP virtual IP was within the correct subnet and not assigned as a physical interface address on any router.

Difficulties were also experienced with DHCP functionality, where clients were unable to obtain IP addresses. This was traced back to incorrect or missing `ip helper-address` configurations on router interfaces. Once the correct DHCP server IP was specified on the appropriate VLAN interfaces, the issue was resolved.

DNS resolution presented challenges across branch networks. While the DNS server functioned correctly within the Darwin site, clients in remote branches were unable to resolve domain names such as *school.com*. This was due to local DNS servers lacking the required records and the limitation of DNS forwarding in Packet Tracer. The problem

was resolved by configuring secondary DNS settings on clients or replicating necessary DNS records across branch DNS servers.

During routing implementation, issues were encountered with OSPF network statements and incorrect area assignments, which prevented proper adjacency formation. Additionally, care had to be taken when implementing route redistribution between EIGRP and OSPF to ensure correct metrics and avoid missing routes.

Finally, while configuring Access Control Lists (ACLs), syntax errors and incorrect rule ordering initially caused unintended traffic behavior. This was resolved by restructuring the ACL with proper permit and deny logic, ensuring that student traffic was appropriately controlled while still allowing access to required services such as the web server.

Overall, these challenges were addressed through systematic troubleshooting, including verification of configurations, use of diagnostic commands, and incremental testing, resulting in a fully functional and stable network.